

Graph-Based Analysis of EEG Functional Connectivity for Eyes-Open and Eyes-Closed Brain State Classification

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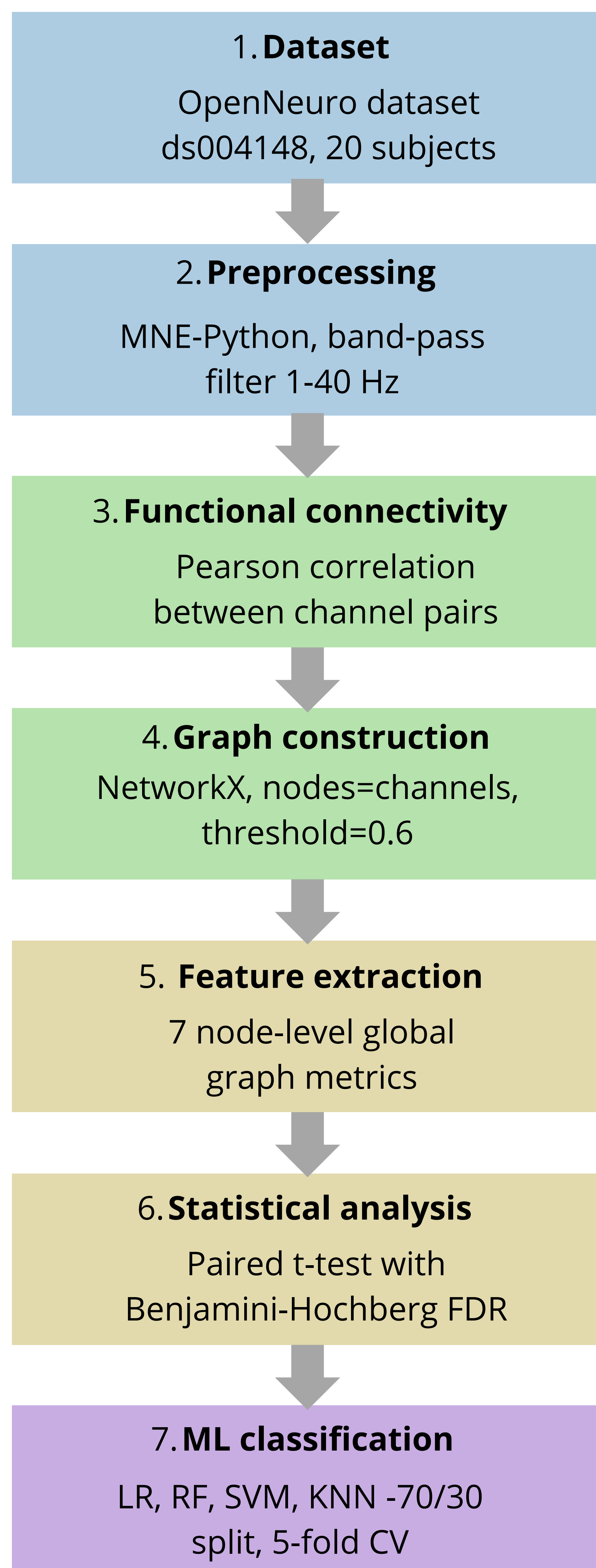
INTRODUCTION

EEG measures brain activity¹. Functional connectivity models interactions between brain regions as networks, and graph theory extracts quantitative features that can be used for brain state classification²⁻⁴.

OBJETIVE

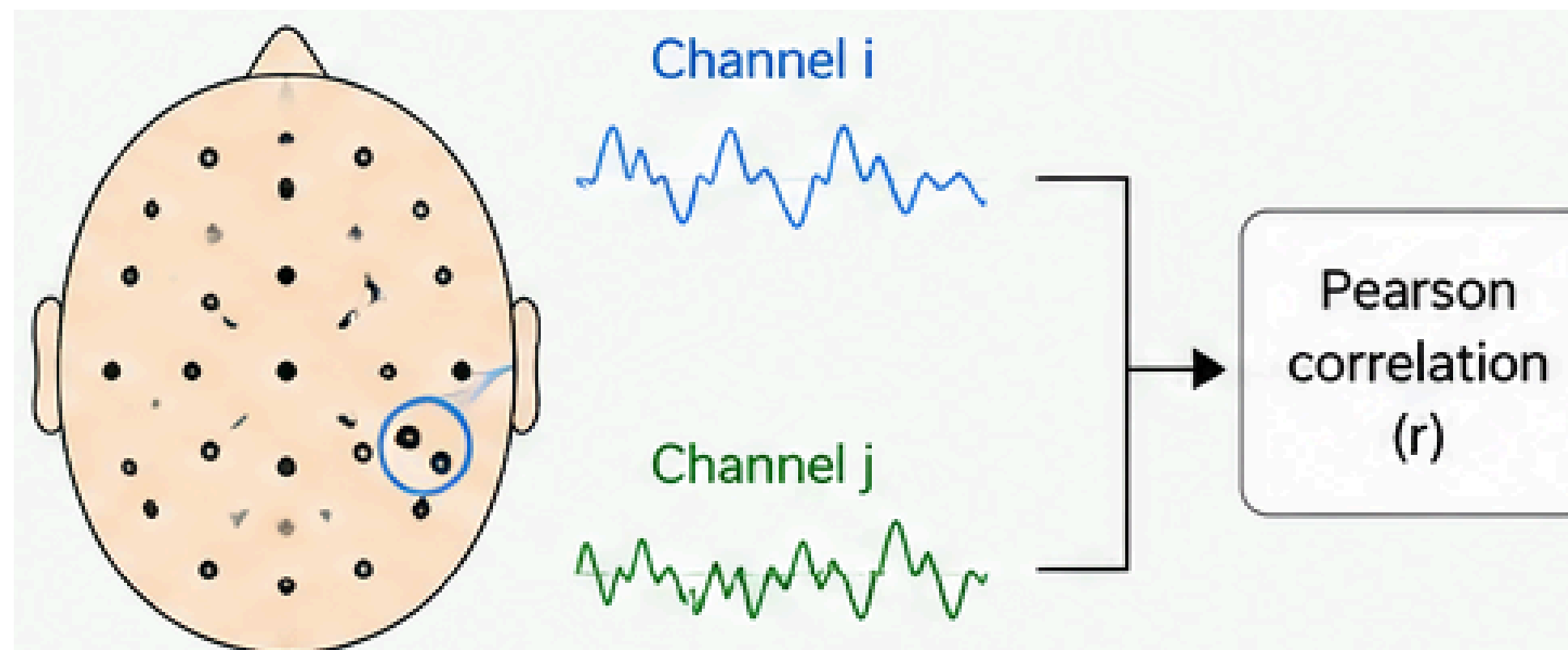
To evaluate whether graph-theoretical features extracted from EEG functional connectivity can discriminate eyes-open and eyes-closed brain states using machine learning.

METHODOLOGY



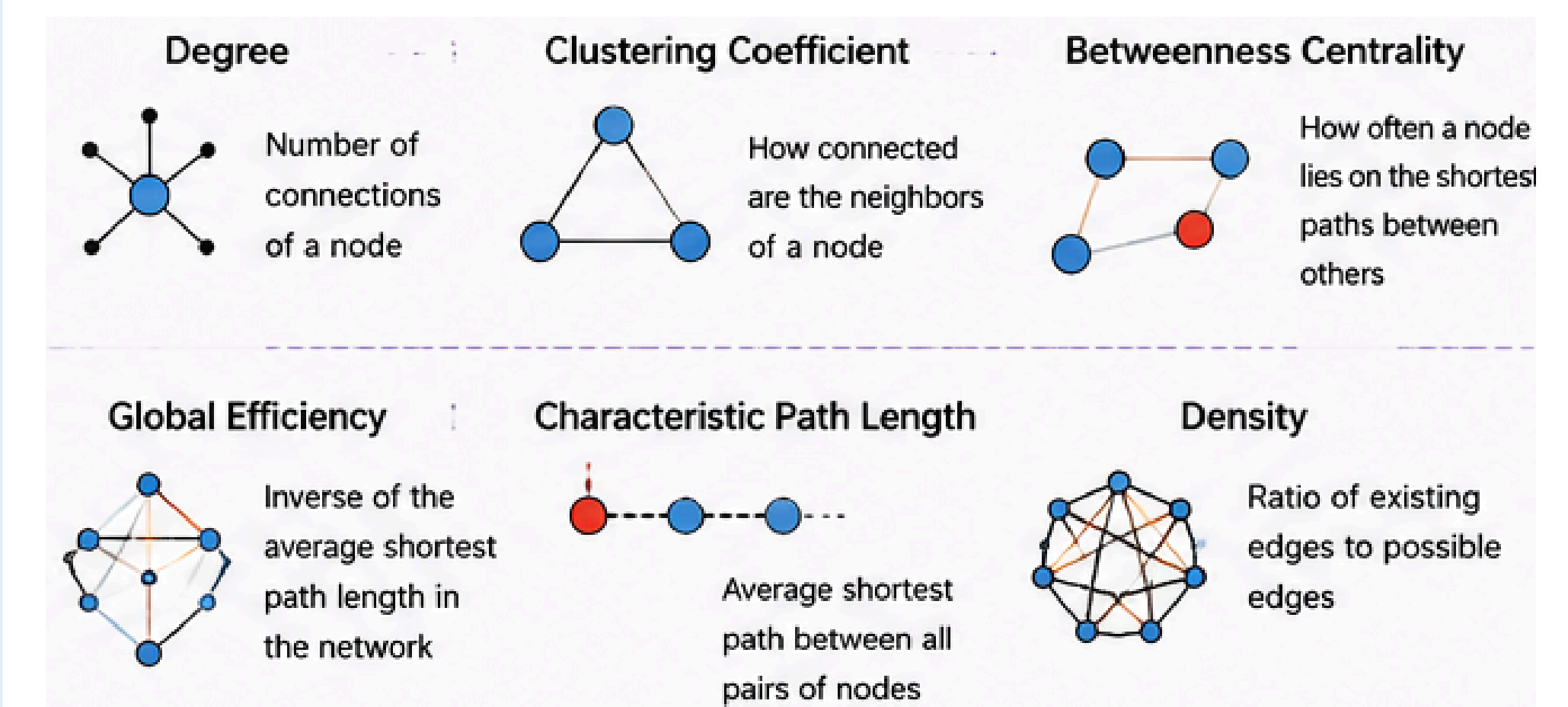
FUNCTIONAL CONNECTIVITY

Describes the statistical relationship between the activity of different brain regions.



GRAPH THEORY

Represents the brain as a network, where nodes correspond to EEG electrodes and edges represent functional connections.



RESULTS

Table 1. Paired t-test with FDR correction for graph-theoretical metrics

Metric	t_stat	p_value	p_FDR	significant FDR
Assortativity	-4.054	0.001	0.005	TRUE
Mean Degree	-2.032	0.056	0.066	FALSE
Density	-2.032	0.056	0.066	FALSE
Mean Betweenness	2.116	0.048	0.066	FALSE
Average Path Length	2.179	0.042	0.066	FALSE
Global Efficiency	-2.154	0.044	0.066	FALSE
Mean Clustering	0.210	0.836	0.836	FALSE

Fig. 1. Global graph-theoretical metrics in open and closed eyes EEG conditions

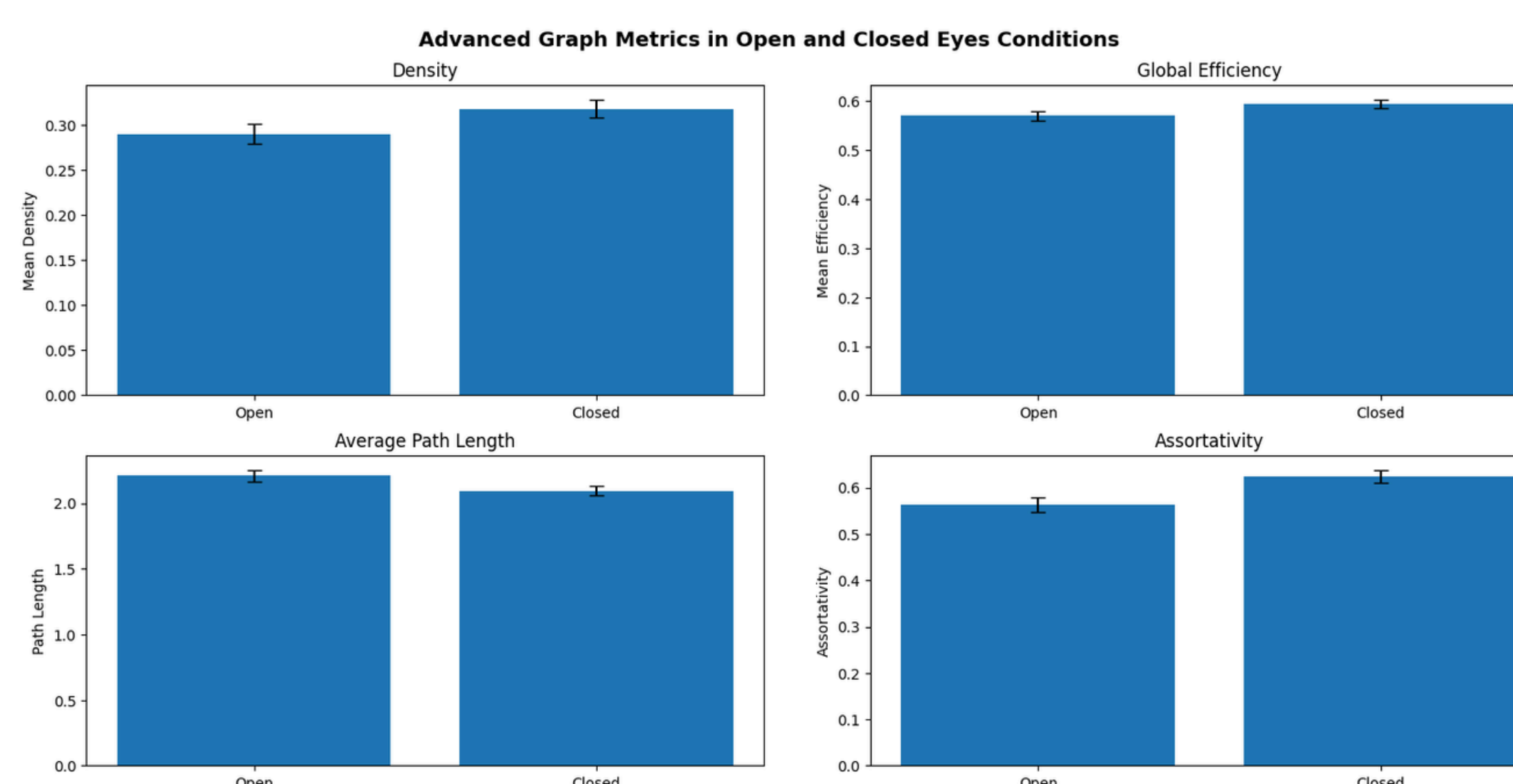


Fig. 2. Local connectivity metrics in open and closed eyes EEG condition

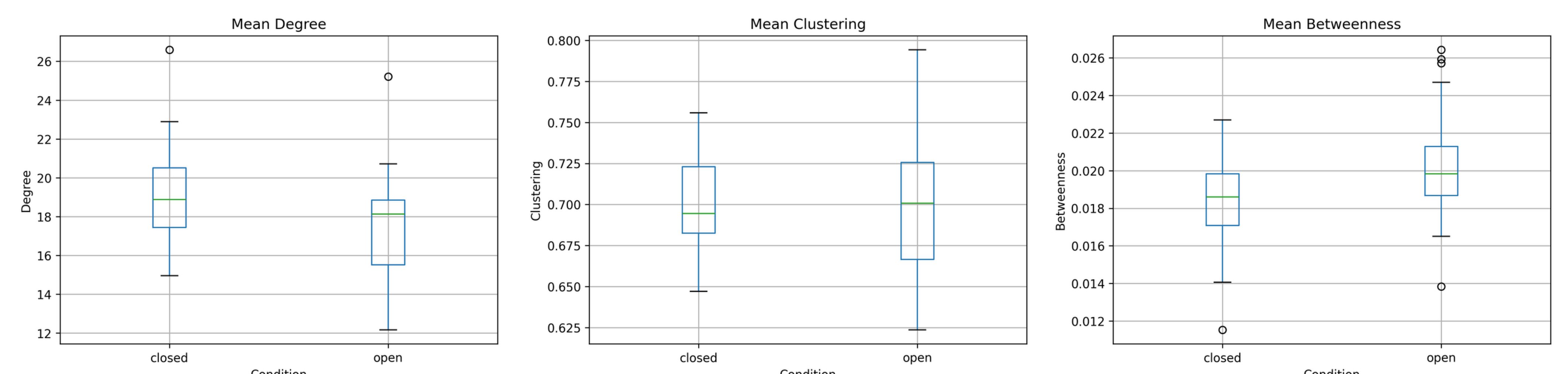


Fig. 3. Cross-validation performance of machine learning models for EEG classification

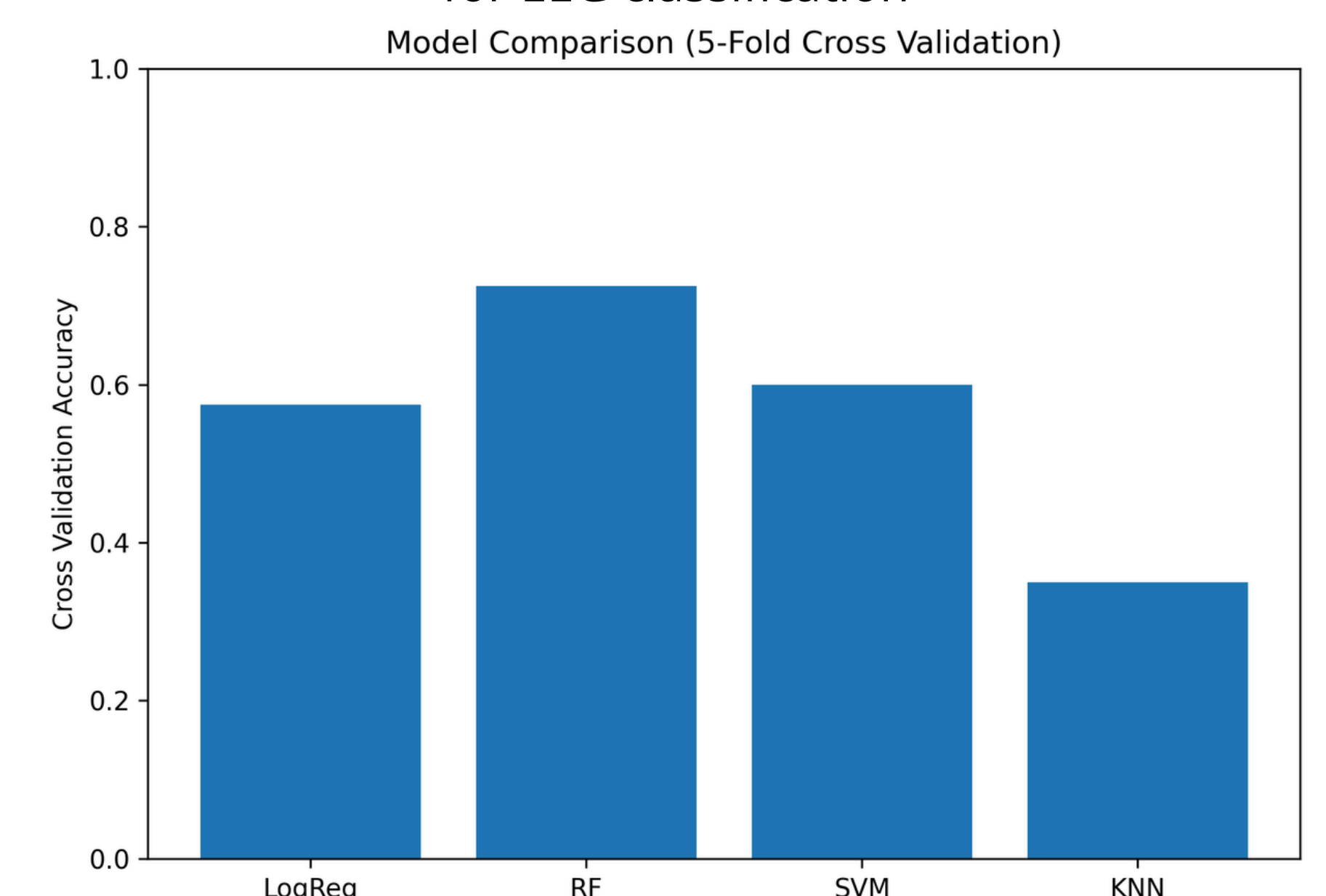


Table 2. Machine learning model performance for classification of EEG conditions

Model	Mean Accuracy	Std
Logistic Regression	0.575	0.061
Random Forest	0.725	0.146
SVM	0.600	0.123
KNN	0.350	0.146

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CONCLUSIONS

- Graph metrics contain discriminative information.
- Assortativity is a sensitive biomarker.
- Random Forest achieved the best performance (72.5%).
- Graph theory provides interpretable features for EEG analysis.